

Apala Pramanik

Lincoln, Nebraska, USA

✉ apamanik2@huskers.unl.edu | 🏠 <https://apalapramanik.github.io/> | 💻 github.com/apalapramanik | 🔗 [linkedin.com/in/apala-pramanik](https://www.linkedin.com/in/apala-pramanik)

Personal Profile

A highly motivated PhD student at the School of Computing, University of Nebraska Lincoln. I am focused on millimeter wave communication improvement strategies using machine learning, with a strong foundation in Robotic Cyber-Physical Systems and formal verification methods. I'm interested in full-time internship opportunities to start in the summer of 2026 in the field of machine learning in wireless communications with applications in V2I/V2X.

Education

University of Nebraska-Lincoln- PhD

Graduate Research Assistant - School of Computing

Advised by: Dr. Mehmet Can Vuran, Dale M. Jensen Chair Professor

Nebraska, USA

June 2024 - Current

University of Nebraska-Lincoln- Master's

Graduate Research Assistant - School of Computing

Advised by: Dr.Dung Hoang Tran, Assistant Professor

Nebraska, USA

Aug 2021 - May 2024

Guru Gobind Singh Indraprastha University- Bachelor's

Electronics and Communication

Thesis: Deep learning techniques for Magnetic Resonance Image (MRI) denoising and reconstruction from undersampled MRI scans.

Delhi, India

Aug 2017 - May 2021

Research Experience

University Of Nebraska-Lincoln

Doctor of Philosophy, Computer Science

- Conducting research on **millimeter-wave (mmWave) communication systems** for next-generation wireless networks, focusing on beamforming, signal processing, and transmit power control with real hardware testbeds.
- Experience with ray-tracing software such as **Wireless InSite** and **Sionna RT**.
- **Technical Skills:** GNU Radio, USRP, Python, C++, Ubuntu Linux, Git.

Nebraska, USA

Aug 2024 - Current

University Of Nebraska-Lincoln

Master's, Computer Science

- Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems
- Experience with **Neural Network Verification (NNV)** software for formal analysis of deep learning models.
- Knowledge of **Signal Temporal Logic (STL)** for specifying and verifying temporal properties in cyber-physical systems.
- **Technical Skills:** ROS, Gazebo, RVIZ, HTML, CSS, Python, C++, Ubuntu Linux, Git.

Nebraska, USA

Aug 2021 - December 2024

Publications

CONFERENCE PROCEEDINGS

Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems

Apala Pramanik, Sung Woo Choi, Yuntao Li, Luan Viet Nguyen, Kyunki Kim, Hoang-Dung Tran

2024 22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE), 2024

ASTITVA: assistive special tools and technologies for inclusion of visually challenged

Apala Pramanik, Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Rohan Tripathi

2021 international conference on computing, communication, and intelligent systems (ICCCIS), 2021

START: Smart Stick based on TLC Algorithm in IoT Network for Visually Challenged Persons

Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Apala Pramanik

2020 Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC), 2020

WORKSHOPS

PerM: Tool for Perception-based Runtime Monitoring for Human-Construction Robot Systems
Accepted at DAC 2024 WIP Workshop
Apala Pramanik, Kyungki Kim, Dung Hoang Tran

Vision-based Runtime Monitoring for Human-Construction Robot Systems
IROS 2023 Workshop : Formal methods techniques in robotics systems: Design and control
Apala Pramanik, Kyungki Kim, Dung Hoang Tran

University Projects

Spanish News Classification
University of Nebraska Lincoln

Nebraska , USA
Aug 2021 - July 2024

- Experimented with several models, including GRU, LSTM and BETO model, we were able to predict our dataset’s labels with a 91.80% accuracy using the BETO model and Tokenizer
- Explored data augmentation techniques such as synonym replacement and character replacement for generating a larger dataset for training leading to 99% training accuracy.
- Technical Skills:** Python with Pandas, matplotlib, Seaborn, Tensorflow, Deep Learning Models
- Soft Skills:** Presentation skills, Leadership, Teamwork, Logical Thinking.

Deep Learning Projects
University of Nebraska Lincoln

Nebraska , USA
Aug 2021 - July 2024

- Classifying Fashion MNIST + CIFAR-100
- Sentiment Analysis with Sequential Models
- Technical Skills:** Python with Pandas, matplotlib, Seaborn, Tensorflow, Deep Learning Models
- Soft Skills:** Report writing, Logical Thinking, Critical Thinking.

Review

IEEE Transactions on Wireless Communications 2026
IEEE Transactions on Wireless Communications 2025

Involvement

Undergrad Research Fair
K-12 Outreach Fair
Mentorship

Presented research projects to undergraduate computer science major students
Demonstrated research activities to K–12 students
Mentored undergraduate researchers on experimental design, data analysis, and research communication

Skills

Programming
Miscellaneous
Soft Skills

Python: Pandas, PyTorch, Tensorflow, NumPy, Scikit-learn, etc.
ROS: rospy, roscpp
Softwares:GNURadio, Wireless Insite, Matlab, Gazebo, Rviz, Arduino, AVR Studio, Proteus Professional, KEIL
Point Cloud Library
Linux, Shell (Bash/Zsh), $\LaTeX$ (Overleaf/R Markdown), Microsoft Office, Git.
Mentoring, Time Management, Teamwork, Problem-solving, Documentation, Engaging Presentation.

Interests

Paper Review
Technical Writing
Dancing
Fitness

I enjoy critically reviewing research papers to evaluate technical rigor, clarity of methodology, and the strength of scientific contributions.
I enjoy writing research papers to convey my research work in easy and understandable language.
I have a senior diploma degree in Bharatnatyam (Indian Classical Dance) and have training of more than 10 years.
I am passionate about weight training and maintaining a healthy and active lifestyle.

References available upon request.